

Identifying exemplary sustainable cropping systems using a positive deviance approach: Wheat-maize double cropping in the North China Plain

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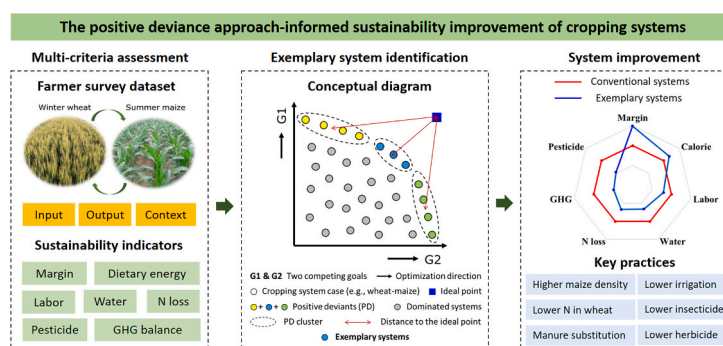
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HIGHLIGHTS

- A positive deviance approach was used to identify exemplary sustainable cropping systems.
- Exemplary wheat-maize systems reached the best compromise across seven indicators.
- Exemplary wheat-maize systems were supported by several efficient practices but not farming contexts.
- Sustainability gaps could be closed if local wheat-maize systems resembled exemplary ones.
- The positive deviance approach is a promising way to inform sustainability improvement of cropping systems.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Sustainable cropping systems need to balance productivity and profitability with resource and environmental conservation. Within a population of cropping system observations, there might be positive deviants that outperform others in terms of sustainability, which could serve as “model systems” for future development. Wheat-maize double cropping is the dominant system in the North China Plain, which is facing multiple economic, societal, and environmental sustainability challenges. Identifying exemplary positive deviants out of a multitude of wheat-maize observations might provide solutions to enhance overall sustainability. **OBJECTIVES:** We aimed to 1) identify exemplary wheat-maize systems that reached optimal performance across seven sustainability indicators, 2) determine which factors regarding management practices and farming contexts resulted in the sustainability gaps between exemplary and other systems, and 3) propose a sustainable wheat-maize prototype. **METHODS:** Based on a farmer survey dataset ($n = 344$), we developed a cropping system-level positive deviance approach, including multi-criteria assessment, positive deviant identification (Pareto ranking) and positive

Abbreviations: NCP, North China Plain; N, Nitrogen; GHG, Greenhouse gas; C, Carbon; MIDIP, Multi-indicator distance to the ideal point; HDIP, Heterogeneity of distances to the ideal point across the indicator set.

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deviant clustering, to identify exemplary wheat-maize systems. We then compared exemplary and other systems to quantify the sustainability gaps and identify the key variables explaining sustainability gaps.

RESULTS AND CONCLUSIONS: Sixteen percent of wheat-maize cases were Pareto-optimal and were classified as positive deviants. These were sorted into seven clusters representing contrasting sustainability patterns. Among these clusters, one comprised exemplary systems due to the best compromise over the indicator set. Compared to remaining wheat-maize cases, exemplary systems, on average, resulted in 49% and 17% higher gross margin and dietary energy output, respectively, and 33–51% lower labor use, groundwater depletion, N loss, net greenhouse gas emission, and pesticide use. Key practices conferring exemplary system performance included higher maize seeding density, lower fertilizer N input in wheat, partial substitution of inorganic fertilizer with manure, a smaller number of irrigation events, and a lower frequency of pesticide and herbicide application. No significant difference in farming context was found between exemplary and other systems.

SIGNIFICANCE: Since the practices of exemplary systems were already locally adopted and proven, we expect that farmers in the region can increase the sustainability of their wheat-maize production by adjusting their management to resemble the exemplary systems. The positive deviance approach thus provides a pragmatic bottom-up approach to identify practices that can improve the sustainability of cropping systems, and can be used for other cropping systems elsewhere.

1. Introduction

With a growing world population and widespread agriculture-induced environmental harms, global crop production must meet joint sustainability demands regarding food security, resource conservation, environmental soundness, and socioeconomic viability (Foley et al., 2011). Therefore, most agricultural regions on the globe need to explore suitable pathways to improve multi-dimensional sustainability of crop production (Lechenet et al., 2017; Spiegel et al., 2018; Davis et al., 2019), although the challenges they face differ among regions (Zhang et al., 2021).

The North China Plain (NCP) is the breadbasket of China, responsible for 59% and 26% of the national wheat and maize production, respectively (National Bureau of Statistics of China (NBSC), 2020). Agriculture in this region has prioritized high productivity to support increasing domestic consumption in the past decades (Niu et al., 2021). The average grain yield increased from 2.0 to 6.4 Mg/ha for wheat (*Triticum aestivum* L.) and from 3.7 to 6.1 Mg/ha for maize (*Zea mays* L.) between 1980 and 2019, with substantial increases in agricultural inputs, such as water, fertilizer, and pesticides (National Bureau of Statistics of China (NBSC), 2020). However, local cropping systems are currently facing a range of sustainability challenges from economic, societal, and environmental perspectives. On the one hand, improving productivity and profitability are needed to ensure national food self-sufficiency and boost rural economic prosperity under the condition of rural labor shortage (Ghose, 2014; Xu et al., 2019). On the other hand, due to intensive crop management by smallholders, crop production has been a major force behind local environmental deterioration. The NCP is suffering from severe groundwater table decline (at a rate of 1 m per year; Kong et al., 2016), fertilizer nitrogen (N)-related water and atmospheric pollution (Meng et al., 2018a; Wang et al., 2019), high greenhouse gas (GHG) emissions (Xu and Lan, 2017), as well as pesticide contamination of soil and water (Brauns et al., 2018). All challenges are pressing, and thus local cropping systems need to pursue multi-dimensional sustainability.

Double cropping of wheat and maize is the largest cropping system, covering around 60% of the total arable land area in the NCP (Li and Lei, 2021). Improving multi-dimensional sustainability of the wheat-maize system would therefore make a great contribution to enhancing farming sustainability in this region. There is indeed a huge potential to improve the sustainability of wheat-maize systems. For instance, many field trials in the NCP have proven that the N loss, GHG emissions, or groundwater table decline could be decreased without sacrificing grain output through optimized management techniques such as fertilization based on soil testing, enhanced-efficiency fertilizers, and advanced irrigation regimes (Ju et al., 2009; Sun et al., 2015; Zheng et al., 2020). However, most of those studies considered only a few aspects of sustainability, intentionally or unintentionally overlooking other aspects

and possible trade-offs.

It is difficult for researchers to give pragmatic solutions to enhance all domains of the sustainability performance of cropping systems based on a “top-down” approach. That is because trade-offs between sustainability domains usually occur in agricultural systems (Kanter et al., 2018; Li et al., 2019) and challenges concern multiple disciplines. Moreover, those solutions may not fit the site-specific conditions, thus being not locally viable for farmers (Liu et al., 2018). So far, a holistic sustainability assessment of wheat-maize systems in the NCP is lacking, and it remains unclear how local farmers could reconcile the different domains of sustainability in wheat and maize production.

Cropping systems within an agricultural region tend to have heterogeneous agronomic, environmental as well as socioeconomic performance as a result of varying cultivation practices and different production contexts across farmers (Aouadi et al., 2015). Identifying the existing cropping systems that are cultivated sustainably from the multitude of observations might provide a suitable example for wider application.

The positive deviance approach can be used to investigate the characteristics and practices of the best performing individuals (i.e., positive deviants) within a population (Herington and van de Fliet, 2018). Here, positive deviants refer to the cases achieving superior performance compared to others within a community despite sharing similar environment, resources, and barriers. The positive deviance approach was initiated and widely applied in the field of human nutrition and public health (Albanna and Heeks, 2019), for example, to propose dietary interventions against child malnutrition through studying positive deviant households with healthier children under given resource-poor circumstances. This approach was also extended to the agricultural domain to look into positive deviant farms and the practices driving them (Savikurki, 2013; Dumont et al., 2020).

Recent studies on positive deviance of farming systems identified positive deviants in terms of multiple assessment indicators through systematic and quantitative methods, e.g., Pareto optimality (Modernel et al., 2018; Steinke et al., 2019; Adelhart Toorop et al., 2020). These studies also captured the diversity of positive deviants, and ruled out positive deviants that underperformed in certain indicators using benchmark values for each indicator. Benchmark values can be recognized as threshold values (Modernel et al., 2018) or the population mean could also be used as a benchmark (Adelhart Toorop et al., 2020). However, the use of such benchmarks is to some extent arbitrary, and it would be interesting to pursue an approach to select positive deviants that is benchmark-free. Besides, most of the agriculture-oriented studies focused on the farm scale, while cropping system-level positive deviance analysis to inform multifaceted sustainability improvement of crop production is still understudied.

In this study, we hypothesized that within a population of wheat-maize double cropping cases, there are likely to be exemplary positive

deviants that could reach comparatively superior and well-balanced performance across all sustainability domains, supported by a set of efficient management practices and possibly also by favorable farming contexts. Here we applied a nested positive deviance approach to wheat-maize observations within a county in the NCP. The research objectives were 1) to identify exemplary positive deviant wheat-maize systems without resorting to the use of benchmarks, 2) to identify whether multi-dimensional sustainability gaps between exemplary and other systems were due to differences in management or farming context, and 3) to propose a sustainable wheat-maize prototype based on exemplary systems.

2. Materials and methods

2.1. Methodological framework for the nested positive deviance approach

We used a four-step methodological approach to identify positive deviants and exemplary cropping systems and to extract measures that improve the sustainability of cropping systems (Fig. 1).

STEP 1: Data collection and multi-criteria assessment.

A field survey was conducted to collect data regarding production input-output as well as the biophysical and socioeconomic context of cropping systems (see Section 2.3). Then we used the dataset to quantify cropping system-level performance in terms of indicators related to local sustainability challenges (see Section 2.4).

STEP 2: Identification of positive deviants and exemplary systems.

Positive deviants were identified by selecting Pareto-optimal cropping systems (explained in Section 2.5.1) from the set based on indicator performance. Next, hierarchical clustering was used to divide all positive deviants into several clusters with contrasting sustainability patterns. Subsequently, the positive deviant cluster with the shortest distance to the ideal point (see Section 2.5.2) was selected as the cluster containing those exemplary systems representing the best compromise across the whole indicator set.

STEP 3: Quantification and explanation of sustainability gaps.

We compared sustainability indicators between exemplary and other systems to calculate multi-dimensional sustainability gaps. Then, we used a range of explanatory variables describing crop management, biophysical and socioeconomic context to analyze factors associated with sustainability gaps.

STEP 4: Establishment of a sustainable cropping system prototype.

Finally, based on the above three steps, a sustainable prototype was built, which summarizes and describes crop management and expected sustainability outcomes of exemplary systems. This locally viable prototype could act as a model for local stakeholders on possibilities to advance the sustainability of wheat-maize production.

2.2. Study area description

Quzhou is a typical agricultural county in the NCP (114°50'E–115°13'E, 36°35'N–36°57'N, located in Hebei Province of China), and was chosen as the study area (Fig. S1). Its representativeness for crop production over the NCP can be summarized in the following points:

- 1) **Pedo-climatic circumstances:** Quzhou county has a warm-temperate and monsoonal climate that is typical for the NCP. The mean annual temperature is 13.1 °C with hot summers and cold winters, while the mean annual precipitation is 500 mm, 60% of which occurs from late June to late October. Agricultural soils are calcareous alluvial, characterized by loamy texture, high soil pH (> 7.5), and low soil organic matter content (< 2% in most croplands).
- 2) **Cropping patterns:** Similar to most agricultural counties in the NCP, croplands in Quzhou are dominated by double cropping of wheat and maize, in which wheat is sown in October and harvested in June the next year while maize is sown immediately after wheat harvest in June and harvested in October. Most of the wheat-maize systems are managed by smallholders who have an average farm size of 0.6 ha.

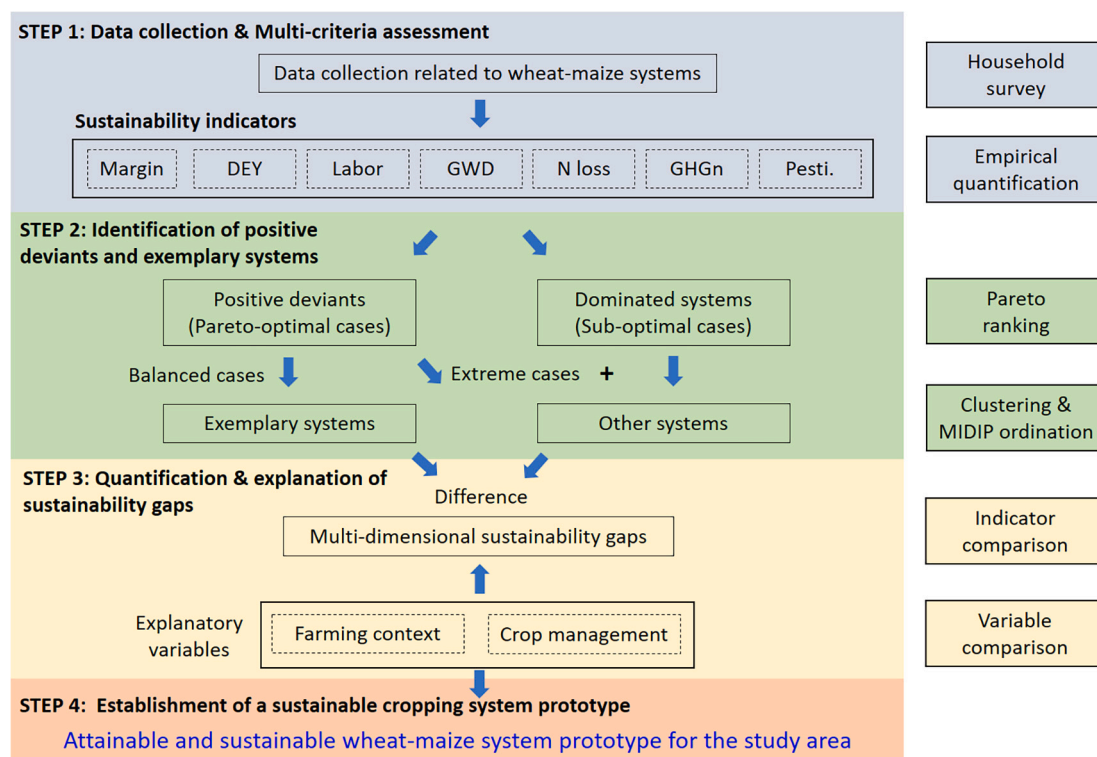


Fig. 1. Schematic diagram of the nested steps in the methodological framework. Boxes outside the flow chart indicate the methods used in the corresponding steps. Margin, DEY, GWD, GHGn, and Pesti. denote gross margin, dietary energy yield, groundwater depletion, net GHG emission, and pesticide use respectively. MIDIP is the multi-indicator distance to the ideal point (see Section 2.5.2).

3) Sustainability challenges: Quzhou faces a variety of economic, societal, and environmental challenges that are characteristic for the NCP: low farming income, rural labor shortage, groundwater table decline, and high N loss (Zhang et al., 2016; Xu et al., 2018; Meng et al., 2021; Feng et al., 2022). Besides, current agri-environmental policies aim to reduce the currently high use of pesticides and high emissions of GHG (Ministry of Agriculture and Rural Affairs of the People's Republic of China (MARA), 2022).

2.3. Farm household survey and dataset establishment

In July 2020, we conducted a survey of 525 farming households in Quzhou. The survey was carried out in 25 villages, with 15 randomly selected households per village (Fig. S1). For each household, we interviewed the household head in person with a structured questionnaire to collect data on the household situation (e.g., labor availability), cultivated cropping systems (e.g., wheat-maize double cropping), and inputs and outputs (e.g., fertilizer use and yield) of each crop species in a one-year cropping cycle (June 2019 – June 2020). We found that 344 households (66%) cultivated wheat-maize double cropping with irrigation from groundwater. Data associated with these wheat-maize observations, including a crop input-output inventory and household characteristics (Table S1), were extracted to build a new dataset for this study. We used the latest soil grid dataset of Quzhou (sampling in 2018; Li et al., 2022) to get averaged soil properties at the village level using the inverse distance weighted method in ArcMap 10.8 (<https://www.esri.com/en-us/arcgis/products/arcgis-desktop>). Eventually, we had a dataset containing variables on crop input, crop output, as well as the biophysical and socioeconomic context (i.e., soil properties and household characteristics) of all groundwater-irrigated wheat-maize cases in Quzhou.

2.4. Multi-criteria assessment

In this study, sustainability indicators were selected accounting for local challenges to be tackled and can fall into three domains, i.e., economic, societal, and environmental aspects, of sustainability. The temporal scale of all indicators is a double-cropping cycle (one year).

2.4.1. Economic domain – Gross margin

Maximizing the profitability of the cropping system is the major interest of farmers. Here annual gross margin (GM_{total} , USD/ha/year) was used to indicate the economic performance of each wheat-maize observation, and was calculated as follows:

$$GM_{\text{total}} = \sum_i Y_i \times PR_i - VC_i \quad (1)$$

where i (= wheat, maize) denotes crop species. Y_i (Mg/ha) and PR_i (USD/t) denote grain yield and farm gate price respectively. Variable costs (VC_i , USD/ha/year) cover all costs of agricultural material inputs (including seed, fertilizer, manure, electricity, pesticides, and growth regulators) and hired machinery use (for tillage, sowing, harvesting, and straw return) based on their application rate or the number of applications and corresponding prices. Prices are listed in Table S2.

2.4.2. Societal domain

2.4.2.1. Dietary energy yield. Dietary energy yield (DEY_{total} , GCal/ha/year) is a widely-used proxy indicator for food provisioning capacity, which is associated with the societal demand for food security. Here, this indicator represents the summation of the energy output of wheat and maize, and was estimated based on Y_i and dietary energy content (E_i , Gcal/Mg) of harvested commodities:

$$DEY_{\text{total}} = \sum_i Y_i \times E_i \quad (2)$$

where dietary energy content of wheat and maize grain was 3.17 and 3.35 GCal/Mg, respectively (Yang et al., 2018).

2.4.2.2. Labor use. Given the increasing rural labor shortage in the NCP, labor input needs to be minimized in sustainable cropping systems. Here we summed up the time farmers spent on manual cropping activities for wheat-maize systems to quantify annual labor use (LU_{total} , h/ha/year), which was expressed as:

$$LU_{\text{total}} = \sum_i \sum_j L_{j,i} \times N_{j,i} \quad (3)$$

where j (=1, 2, 3, ...) represents a manual cropping activity including mineral fertilizer and manure application, pesticide and herbicide use, irrigation, weeding, and harvesting. $L_{j,i}$ is labor time consumption for activity j , and $N_{j,i}$ denotes the number of times the activity j was performed during the growing season of crop species i . Because some farmers could not recall their labor inputs precisely and $L_{j,i}$ was found to be relatively homogeneous among households, we used a fixed value for $L_{j,i}$ (except for manure use and weeding), equal to the average of all households able to report it (Table S3).

2.4.3. Environmental domain

2.4.3.1. Groundwater depletion. Sustainable wheat-maize systems should contribute to mitigating groundwater table decline in the NCP. Groundwater depletion (GWD, mm/year) is a useful indicator to assess the sustainability of groundwater use at the cropping system level (Yang et al., 2021), and was defined as:

$$GWD = \sum_i IR_i - D \quad (4)$$

where IR_i (mm) is the irrigation amount from groundwater for crop i , and D (mm/year) is the deep infiltration during a wheat-maize cropping year. Positive values of GWD reflect groundwater depletion, whereas negative values indicate groundwater recharge. IR_i was estimated from the electricity consumption of water pumping using an empirical equation (Section 1.1 in the Supplemental Materials) while D was calculated as:

$$D = \alpha \times \left(P + \sum_i IR_i \right) \quad (5)$$

where P is annual precipitation (mm/year). α denotes the dimensionless recharge coefficient which was set as 0.2 calibrated for the case study area (Yang et al., 2021).

2.4.3.2. Nitrogen loss. Fertilizer nitrogen (N) that is lost to the environment causes water eutrophication and PM2.5 (fine particle matters) pollution (Huang et al., 2017; Meng et al., 2018a). Therefore, N loss (NL_{total} , kg N/ha/year) from wheat-maize croplands needs to be minimized and was defined as:

$$NL_{\text{total}} = \sum_i (NL_{\text{NH}_3,i} + NL_{\text{NO}_3,i} + NL_{\text{N}_2\text{O},i}) \quad (6)$$

where $NL_{\text{NH}_3,i}$, $NL_{\text{NO}_3,i}$ and $NL_{\text{N}_2\text{O},i}$ indicate the crop-specific amount of N (kg N/ha) lost via the pathways of ammonia volatilization, nitrate leaching, and nitrous oxide emission, respectively. These were calculated based on N fertilizer inputs using empirical equations calibrated for the NCP (Section 1.2 in the Supplemental Materials).

2.4.3.3. Net GHG emission. Carbon (C) emission abatement has been recognized as a key objective for crop production in China (Ministry of Agriculture and Rural Affairs of the People's Republic of China (MARA), 2022). Building on the idea of life cycle assessment, net GHG emission

was used to evaluate the system performance in combating global warming (International Organization for Standardization (ISO), 2006). The system boundary included manufacturing and transportation of agricultural inputs, cropping practices on farmland, and soil carbon stock change (including pre-and on-farm stage). Net GHG emission (GHG_{net} , $\text{Mg CO}_2 \text{ eq./ha/year}$) depends on GHG emission (GHG_{emi} , $\text{Mg CO}_2 \text{ eq./ha/year}$) and soil C sequestration (Q_{SOC} , $\text{Mg CO}_2 \text{ eq./ha/year}$):

$$\text{GHG}_{\text{net}} = \text{GHG}_{\text{emi}} - \text{Q}_{\text{SOC}} \quad (7)$$

where GHG_{emi} accounts for pre-farm GHG emissions, N fertilizer-induced direct and indirect nitrous oxide emission, and soil C degradation, while Q_{SOC} considered C inputs from straw return, root residues, and animal manure. These were all quantified using empirical equations specified in Sections 1.3–1.4 of the Supplemental Materials.

2.4.3.4. Pesticide use. Less reliance on chemical pesticides is an important element of sustainable cropping systems. Pesticides included insecticides, herbicides, and fungicides in our research. Because it was difficult for farmers to report the exact product composition and active ingredients applied, we quantified the pesticide inputs based on the number of applications of a full dosage according to label recommendation (usually a 20 L tank of knapsack sprayer per mu, 1/15 ha, one tank as one dose). Thus, pesticide use (PU_{total} , doses/year) was calculated as:

$$\text{PU}_{\text{total}} = \text{IU}_{\text{total}} + \text{HU}_{\text{total}} + \text{FU}_{\text{total}} \quad (8)$$

where IU_{total} , HU_{total} , and FU_{total} indicate insecticide use, herbicide use, and fungicide use in a one-year cropping cycle (all expressed in doses/year), respectively. The calculation is further elaborated in Section 1.5 of the Supplemental Materials.

2.5. Identification of positive deviants and exemplary systems

This step aims to select exemplary systems out of all wheat-maize observations in Quzhou in terms of the above sustainability indicators, which can be regarded as the “model” for sustainable wheat-maize production (Fig. 2).

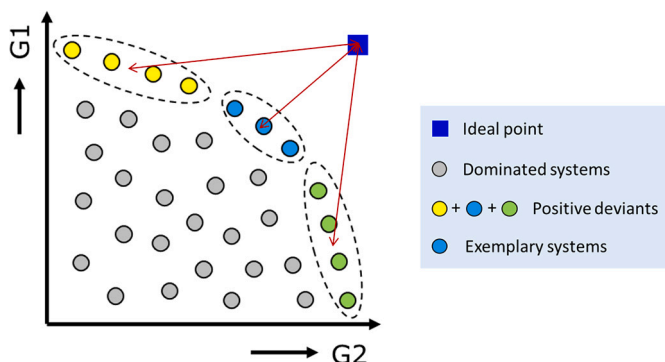


Fig. 2. Visual representation of positive deviants (circles filled with yellow, light blue, and green), dominated systems (circles filled with grey), and exemplary systems (circles filled with light blue). Each circle represents a cropping system observation. G1 and G2 denote two competing goals of systems, and the solid arrows indicate the respective optimization directions. Each ellipse with a dashed border denotes a positive deviant cluster. The blue square is the ideal system, corresponding in all indicators with the best value observed over the whole population of cases. The red arrows indicate the multi-indicator distance from the cluster centroid to the ideal system (MIDIP). Positive deviants, also known as Pareto-optimal systems, are not dominated by any other systems. Exemplary systems are defined as the positive deviants that reach the best compromise between multiple goals and their cluster is characterized by the shortest MIDIP. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.5.1. Pareto ranking-based selection

Pareto ranking is a method that can find out well-performing individuals, also known as Pareto-optimality, from the whole population without assigning subjective weights to multiple indicators (Groot and Rossing, 2011). Here we used Pareto rank as a criterion to identify positive deviants among wheat-maize cases, in terms of seven sustainability indicators (Fig. 1). During the ranking process, cases were ranked as “1” (i.e., Pareto-optimal or non-dominated) when they are not inferior than any other case for all indicators (Groot et al., 2012). The Pareto ranking was performed by using *pareto_rank* function from R package “eaf”. In this study, we treated Pareto-rank-1 cases as positive deviants, while the remaining systems were referred to as dominated systems.

2.5.2. Clustering- and distance-based selection

In the set of positive deviants, there might be extreme cases that excelled at certain indicators but still performed relatively poorly in other indicators, in case strong trade-offs among indicators exist (Adelhart Toorop et al., 2020). When facing multiple challenges without predefined priority, we need to identify the positive deviants that reach an acceptable compromise across the whole indicator set. Such positive deviants were denoted as “exemplary systems”, while the rest of the positive deviants and all dominated systems together as “other systems”. An additional method is needed to identify exemplary systems.

We applied a hierarchical clustering method, with seven normalized indicators being response variables, to partition all positive deviants into several clusters with contrasting characteristics of sustainability performance. This clustering method used Manhattan distance, which is not sensitive to outliers, to measure dissimilarities among cases, and used the “Ward.D2” linkage approach to minimize the intra-cluster dissimilarities (Kassambara, 2017). We conducted the clustering using the function *hclust* in R software. The decision on the number of clusters should be made to keep intra-cluster variation in indicator performance small enough, which enabled each of the clusters to be treated as a cropping system group with a certain sustainability pattern. Here, we made an elbow plot using function *fviz_nbclust*, and determined the number of clusters based on the position of the bend of the elbow plot. Then positive deviants were assigned to the clusters using function *cutree*.

Subsequently, to evaluate the degree of compromise across indicators, two metrics were applied for each positive deviant cluster: multi-indicator distance to the ideal point (MIDIP) and heterogeneity of distances to the ideal point across the indicator set (HDIP). The ideal point was defined as a point in performance space with coordinates that are for each indicator equal to the best observed indicator value in the population of observations. MIDIP was defined as the range-scaled Euclidean distance from the centroid of a given cluster to the ideal point. We then defined the best compromise cluster of positive deviants as the cluster that had the shortest MIDIP in scaled indicator space (each indicator scaled from 0 to 1). This definition of the best compromise gives equal weight to different indicators and is scale-invariant. MIDIP was derived as:

$$\text{MIDIP}_c = \sqrt{\sum_{k=1}^n d_{k,c}^2} \quad (9)$$

$$d_{k,c} = \left| \frac{p_{k,c} - q_k}{md_k} \right| \quad (10)$$

$$md_k = \text{maximum}_k - \text{minimum}_k \quad (11)$$

where c ($=1, 2, \dots, 7$) represents positive deviant clusters, and k ($=1, 2, \dots, n$) denotes the sustainability indicators. For indicator k , $d_{k,c}$ means the range-scaled distance from the centroid of cluster c ($p_{k,c}$) to the ideal point (q_k), while md_k is the maximum discrepancy across the whole population.

In rare occasions, the clusters closest to the ideal point may still

include extreme observations. As an indicator for the occurrence of this situation, HDIP, defined as the standard deviation of distances to the ideal point of all indicators, was also quantified for each cluster:

$$\text{HDIP}_c = \sqrt{\frac{\sum_{k=1}^n (d_{k,c} - \bar{d}_c)^2}{n-1}} \quad (12)$$

where n is the number of indicators. $\bar{d}_c$ denotes the mean of $d_{k,c}$ across the indicator set for cluster c . The cluster of exemplary systems should also have the relatively low HDIP to ensure the most balanced performance. When a cluster has the lowest MIDIP but not the lowest HDIP, it is needed to explicitly list $d_{k,c}$ for each indicator of all clusters. This can help to decide whether such a high HDIP is acceptable or not.

2.6. Quantification and explanation of sustainability gaps

2.6.1. Comparison between exemplary and other systems

The sustainability gap for the indicator k (SG_k) was defined as the difference between the mean performance of exemplary systems and that of other systems, expressed as:

$$\text{SG}_k = \text{MP}_{\text{exem},k} - \text{MP}_{\text{other},k} \quad (13)$$

where $\text{MP}_{\text{exem},k}$ and $\text{MP}_{\text{other},k}$ represent the mean values of indicator k of exemplary and other systems, respectively.

In addition to the aforementioned individual sustainability indicators, we also compared the exemplary and other systems in terms of an efficiency indicator (economic labor productivity, ELP, USD/caput) and three footprint metrics including N footprint (NF, kg N/GCal), C footprint (CF, Mg CO₂ eq./GCal), and blue water footprint (BWF, mm/GCal). These indicators were calculated for each observation as follows:

$$\text{ELP} = \text{GM}_{\text{total}} / \text{LU}_{\text{total}} \quad (14)$$

$$\text{NF} = \text{NL}_{\text{total}} / \text{DEY}_{\text{total}} \quad (15)$$

$$\text{CF} = \text{GHG}_{\text{net}} / \text{DEY}_{\text{total}} \quad (16)$$

$$\text{BWF} = \sum_i \text{IR}_i / \text{DEY}_{\text{total}} \quad (17)$$

The difference in mean values of all indicators between exemplary and other systems was tested through t -tests with equal variances at a significance level of 0.05 using the function $t.test$ in R.

2.6.2. Identification of variables underlying sustainability gaps

After quantification of sustainability gaps between exemplary and other systems, we analyzed which factors might lead to these differences in indicators. In this step, we compared various explanatory variables describing management practices (e.g., sowing density, N application rate, number of irrigation events, etc.), biophysical context (e.g., soil organic matter content, soil bulk density), and socioeconomic context (e.g., educational experience of household heads, household labor availability) between exemplary and other systems (Table S7). Differences in values of such indicators between exemplary and other systems were tested by t -tests.

2.7. Establishment of a sustainable cropping system prototype

Sustainability performance and explanatory variables that characterized exemplary systems were summarized to establish a sustainable wheat-maize production prototype. Values in the prototype were the average of all exemplary cases. Flexible options that featured in some of the exemplary cases were also presented. This prototype could inform relevant stakeholders, including agricultural authorities, extension staff, and other producers, of the sustainability gaps of current wheat-maize systems and associated feasible improvement interventions tailored to

local conditions.

3. Results

3.1. Indicator performance, positive deviants, and dominated systems

We found large variations in each of sustainability indicators over the whole population of wheat-maize observations, with coefficients of variation of 0.12–0.53 (Table 1). This shows that the performance of wheat-maize systems greatly varied among households in Quzhou. Using Pareto ranking, 16% of cases ($n = 56$) were classified as positive deviants (Pareto-optimal cases) with Pareto rank 1 while the remainder was classified as dominated systems (sub-optimal cases). Accordingly, positive deviants reached better performance than dominated systems. For example, the average gross margin and dietary energy yield of positive deviants was 24% and 10% higher than that of dominated systems. Meanwhile, groundwater depletion, N loss, net GHG emission and pesticide use of positive deviants were on average 26%, 32%, 34%, and 13% lower, respectively, than that of dominated systems. However, the coefficient of variation of each indicator of positive deviants was still high (0.14–0.50), which means positive deviants had contrasting patterns of sustainability performance with possible trade-offs between different indicators.

3.2. Characteristics of positive deviant clusters and exemplary system identification

Using hierarchical clustering, we divided the positive deviants into seven clusters: PD1-PD7 (Fig. 3; Fig. S2). The clusters PD1-PD4 had higher average economic and dietary energy output than PD5-PD7 and the dominated systems. PD1-PD3, however, underperformed in terms of labor and environmental indicators. Of these three, PD1 had very high pesticide input (15.4 doses/year) and labor input (388 h/ha/year), while PD2 was characterized by high groundwater depletion (281 mm/year), N loss (137 kg N/ha/year), and net GHG emission (6.1 Mg CO₂ eq./ha/year). PD3 had relatively high groundwater depletion (269 mm/year) and pesticide use (8.6 doses/year). On the other hand, PD5-PD7 resulted in markedly lower average gross margin and dietary energy yield than PD1-PD4 and even the dominated systems, though they performed best on some other indicators, i.e., groundwater depletion of PD5, N loss of PD6 and PD7, and labor use of PD7. By contrast, PD4 achieved the best compromise across all indicators, with the lowest values of MIDIP and HDIP (Fig. 3) among the seven clusters. Therefore, we selected PD4 which contained 9 observations (2.6% of the population) as exemplary systems with cultivation practices that might qualify for adoption by other farmers.

3.3. Sustainability gaps between exemplary and other systems

Comparison of multifaceted indicators between exemplary and other systems shows large sustainability gaps of wheat-maize double cropping in Quzhou (Fig. 4; Table S9). Mean gross margin and dietary energy yield of exemplary systems were 49% and 17% higher than the average of other systems ($p < 0.05$). Meanwhile, relative to other systems, from the environmental perspective, exemplary systems resulted in 31%, 69%, and 46% lower groundwater depletion, net GHG emission, and pesticide use ($p < 0.05$), and they had on average 34% lower N loss ($p = 0.058$). The mean labor consumption of exemplary systems was 17% lower than that of other systems ($p = 0.050$), despite a large variation of this indicator for exemplary systems.

Exemplary systems also had 47% higher economic labor productivity while resulting in 44%, 59%, and 38% lower N footprint, C footprint, and blue water footprint, respectively, when compared with other systems ($p < 0.05$; Fig. 5).

Table 1
Indicator performance of wheat-maize double cropping systems in Quzhou.

Indicator	Unit	All observations (n ^a =344)			Positive deviants (n = 56)			Dominated systems (n = 288)		
		Mean	SD ^b	CV ^c	Mean	SD	CV	Mean	SD	CV
Gross margin	USD/ha/yr ^d	2215	689	0.31	2779	706	0.25	2105	630	0.30
Dietary energy yield	GCal/ha/yr	52.3	6.0	0.12	56.9	7.7	0.14	51.6	5.4	0.10
Labor use	h/ha/yr	253	79	0.31	235	78	0.33	256	78	0.31
Groundwater depletion	mm/yr	283	69	0.24	232	62	0.27	293	66	0.23
N loss	kg N/ha/yr	135	71	0.53	96	23	0.24	142	75	0.52
Net GHG emission	Mg CO ₂ eq./ha/yr	7.40	2.12	0.29	5.18	1.13	0.22	7.83	2.00	0.26
Pesticide use	doses/yr	9.0	4.0	0.44	8.0	4.0	0.50	9.2	4.0	0.43

^a n means sample size.

^b SD means standard deviation.

^c CV means coefficient of variation.

^d yr is the abbreviation of year.

3.4. Explanatory variables underlying exemplary systems

Variables concerning crop management were compared between exemplary and other systems (Fig. 6). First of all, exemplary systems tended to have a higher planting density of maize than other systems (7.8 vs. 6.8 plants/m²; $p < 0.05$). Compared with other systems, exemplary systems had 24% lower annual N application rate, mainly due to a lower N input for wheat (203 vs. 288 kg N/ha/season; $p < 0.05$). Fourteen wheat-maize cases adopted animal manure amendment in this study, but only one case with the lowest total N input among them was identified as an exemplary system (Table S10). The total irrigation amount per year of exemplary systems was 27% lower than that of other systems as a result of the lower number of irrigation events in both wheat (3 vs. 4) and maize (1 vs. 2) (both $p < 0.05$). In addition, exemplary systems applied 43% and 38% fewer doses of insecticides and herbicides than other systems, respectively ($p < 0.05$), while no significant difference in fungicide use existed. Wheat sowing density, phosphorus and potassium inputs, and diesel use were not significantly different between exemplary and other systems.

We also compared variables on the biophysical and socio-economic context between exemplary and other systems (Table 2). Four soil properties including soil organic matter, soil bulk density, soil Olsen-P as well as soil electrical conductivity were not different between exemplary and other systems. Likewise, there were no significant differences in household labor availability, income per capita, and educational and cropping experience of the household head between exemplary and other systems.

3.5. Exemplary system-informed prototype of the wheat-maize system

Based on the above results regarding exemplary systems, an attainable prototype of a sustainable wheat-maize double cropping system was established (Fig. 7). This prototype features a best compromise across economic, societal, and environmental indicators. Compared to the conventional wheat-maize system representing non-exemplary systems, the sustainable prototype was characterized by increased maize planting density, lower N input for wheat, manure amendment without an increase in total N input, and a reduced input of irrigation water and pesticides (especially insecticides and herbicides) for both wheat and maize.

4. Discussion

We found large variability in indicator performance among 344 cases of wheat-maize double cropping in Quzhou county in terms of seven economic, societal, and environmental indicators (Table 1). We identified positive deviants (Pareto-optimality), and then partitioned them into seven clusters (PD1-PD7) with contrasting sustainability patterns (Table 1; Fig. 3; Fig. S2). Nine cases (2.6%) constituting PD4 were categorized as “exemplary systems”, which represented the best

compromise sustainability performance over the indicator set, on average outperforming all other systems ($n = 335$) in each indicator (Fig. 4). We identified the practices that could close the sustainability gaps between exemplary and other systems: higher maize planting density, lower N fertilizer rate for wheat, partial substitution of inorganic fertilizer by manure, lower frequency of irrigation and insecticide and herbicide application (Fig. 5). Finally, we proposed an exemplary system-informed prototype of sustainable wheat-maize production (Fig. 6), which could be adopted by local farmers to reconcile multiple domains of agricultural sustainability.

4.1. Identification of exemplary wheat-maize systems for sustainable production

Heterogeneous sustainability performance within a population of existing wheat-maize systems provided an opportunity to identify the sustainable cases exemplary to others. The “bottom-up” perspective that identified sustainable wheat-maize systems can be more suitable to local contexts and better handle multiple (more than three) indicators, than those identified by the “top-down” approaches, e.g., crop expert recommendation (Xiao et al., 2019) and mechanistic model simulation (Xin and Tao, 2019). This is because the “bottom-up” approaches give solutions that have been implemented under the local context, and trade off different sustainability attributes in a way that is already adopted and proven in reality. We therefore identified exemplary systems through investigating the diversity of wheat-maize cultivation by practical farmers.

This study used a positive deviance approach combining Pareto ranking, positive deviant clustering, and distance-based ordination (MIDIP and HDIP) to select wheat-maize systems that reached the best compromise across seven indicators (i.e., exemplary systems). Pareto ranking identified positive deviants accounting for 16% of all observations. Pareto-optimality may include cases that perform well in some indicators at a high cost of others. When we equally emphasize all sustainability domains, these unbalanced positive deviants are not acceptable and thus need to be ruled out. To do so, we used the clustering technique to divide positive deviants into seven clusters, in each of which wheat-maize cases had similar sustainability patterns. Then MIDIP and HDIP ordination of clusters eventually helped us to identify the cluster (PD4) containing nine exemplary (compromise) systems. Compared to conventional approaches, this method has three advantages: 1) Greater capacity to consider multiple indicators than conventional grouping methods such as quadrant and clustering analysis for all observations (Ha et al., 2015; Chen et al., 2020); 2) The benchmark-free method to filter out unbalanced cases and to select cases reaching the best compromise, unlike threshold values- or population mean-based selection (Modernel et al., 2018; Adelhart Toorop et al., 2020; Jiang et al., 2021); 3) Identified exemplary systems are locally viable, whereas externally developed solutions might not fit the local context.

We defined sustainability gaps in this study as the difference in

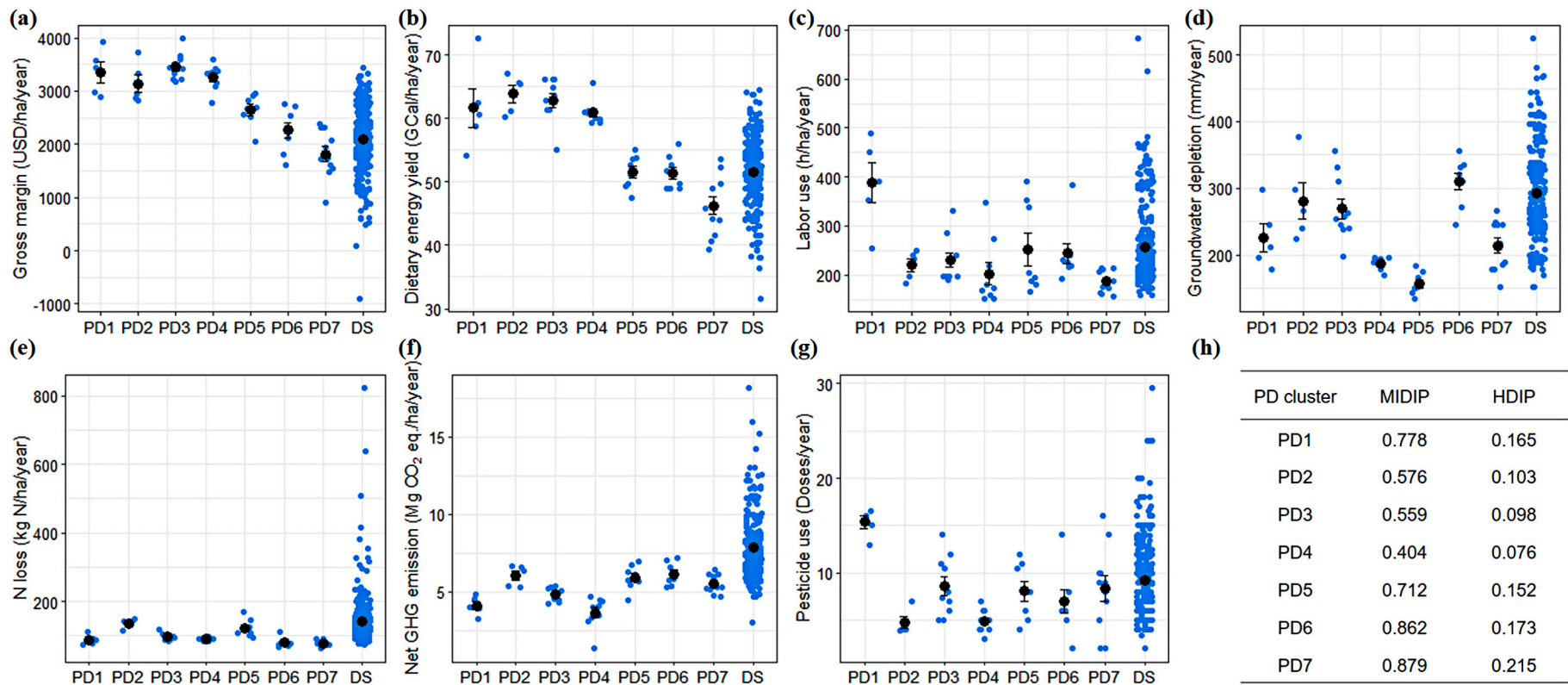


Fig. 3. Indicator performance (a-g) and distance-related outcomes (h) of wheat-maize double cropping systems in Quzhou. All cases were grouped into positive deviant clusters (PD1-PD7) and dominated systems (DS, $n = 288$). A blue dot denotes a wheat-maize observation. The black dot represents the mean of all observations within a cluster, while the error bar represents the standard error. The sample sizes for PD1-PD7 are 5, 5, 10, 9, 8, 8, and 11 respectively. MIDIP indicates the multi-indicator distance to the ideal point, and HDIP indicates the heterogeneity of distances to the ideal point of all indicators. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

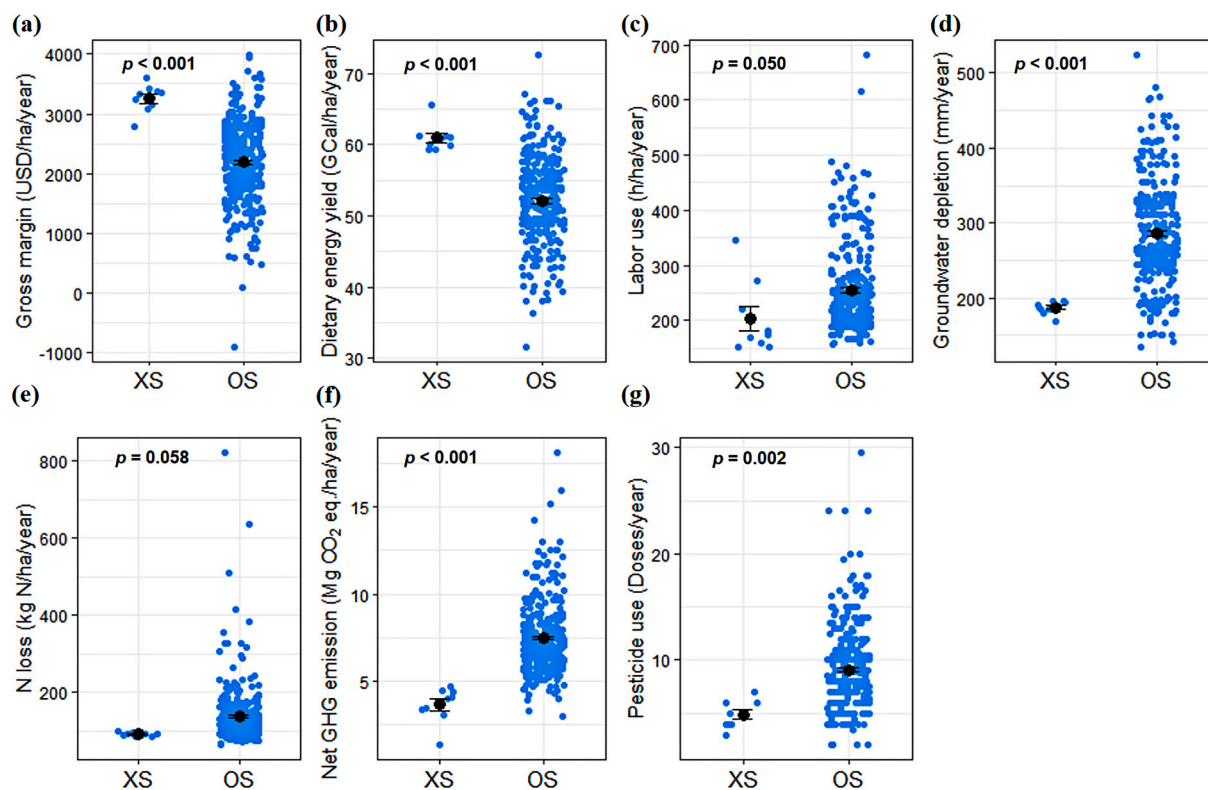


Fig. 4. Sustainability performances (a-g) of the exemplary (XS, $n = 9$) and other systems (OS, $n = 335$) in Quzhou. A blue dot denotes a wheat-maize observation. The black dot represents the mean of all observations in a group while the error bar represents the standard error. The p values are the results of t -tests comparing indicator values of XS and OS. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

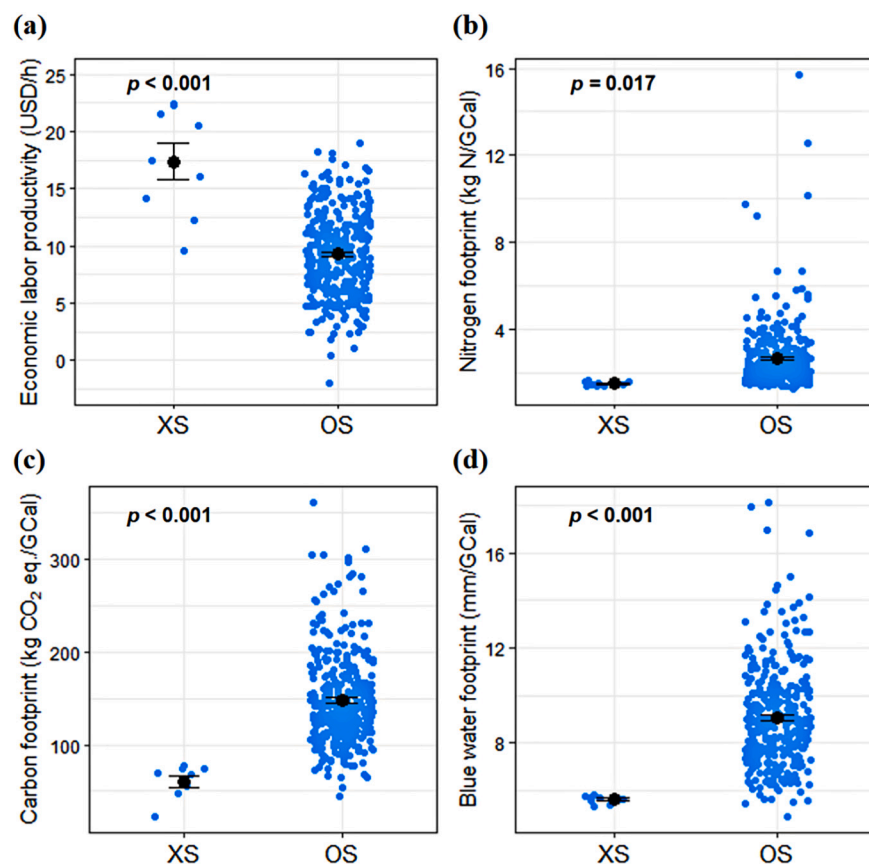


Fig. 5. Economic labor productivity (a) and environmental footprints (b-d) of the exemplary (XS, $n = 9$) and other systems (OS, $n = 335$) in Quzhou. A blue dot denotes a wheat-maize observation. The black dot represents the mean of all observations in a group while the error bar represents the standard error. The p values are the results of t -tests comparing indicator values of XS and OS. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

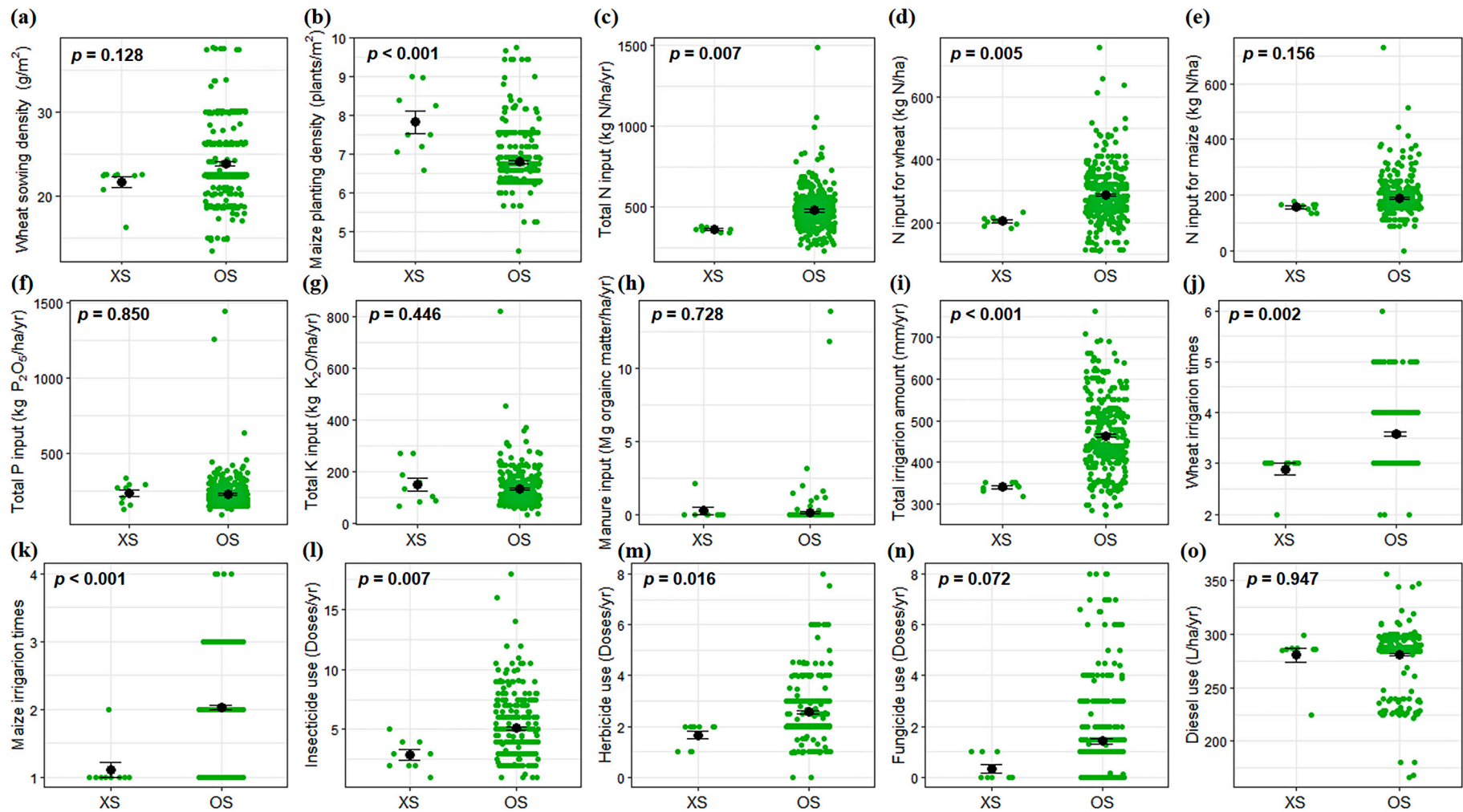


Fig. 6. Variables describing crop management (a-o) of exemplary (XS, $n = 9$) and other systems (OS, $n = 335$) in Quzhou. A green dot denotes a wheat-maize observation. The black dot represents the mean of all observations in a group while the error bar represents the standard error. The p values are the results of t -tests comparing variable values of XS and OS. The yr in the unit is the abbreviation of year. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Variables describing the farming context of the exemplary (XS) and other systems (OS) in Quzhou. Data were shown as “mean $\pm$ standard error of the mean” for each contextual variable. The *p* values are the results of t-tests comparing variable values of XS and OS.

Contextual variable	Unit	XS (n = 9)	OS (n = 335)	<i>p</i> -value
Soil properties:				
Soil bulk density	g/cm ³	1.07 $\pm$ 0.04	1.11 $\pm$ 0.00	0.22
Soil organic matter	g/kg	12.4 $\pm$ 0.7	13.9 $\pm$ 0.2	0.08
Soil Olsen-P	mg/kg	21.6 $\pm$ 1.7	20.6 $\pm$ 0.3	0.61
Soil EC ^a	dS/m	4.36 $\pm$ 0.25	4.18 $\pm$ 0.04	0.48
Household background:				
Labor availability	capita/ha	3.16 $\pm$ 0.83	3.21 $\pm$ 0.15	0.96
	USD/capita/yr	3109 $\pm$		
Income per capita	yr	1546	2199 $\pm$ 134	0.28
Education years	years	6.7 $\pm$ 1.4	5.9 $\pm$ 0.2	0.57
Cropping experience	years	39 $\pm$ 2	37 $\pm$ 1	0.67

^a EC: Electrical conductivity.

indicator performance between exemplary and other systems, representing the attainable scope of improvement using the best compromise over the indicator set as an exemplary system for adoption. The presence of large sustainability gaps in our study indicates that if local wheat-maize cases could adopt practices of exemplary systems, the holistic sustainability of wheat-maize systems in Quzhou could greatly improve. The positive deviance approach is a promising data-driven methodology to inform multi-dimensional sustainability of cropping systems.

4.2. Explanatory variables behind sustainability gaps

Sustainability performance of cropping systems is generally dependent on management practices and farming context (Deytieu et al., 2016). We selected management and contextual variables that are commonly used to characterize wheat-maize systems (Table S7), and identified key variables that might be associated with the sustainability gaps between exemplary and other systems. The results indicate that exemplary systems were mainly underpinned by a number of more sustainable practices but not by more favorable contexts (Fig. 6; Table 2).

Exemplary systems were characterized by a higher maize planting density (7.8 plants/m²). Previous studies showed that a planting density of 7.5–9 plants/m² was crucial for maize to better utilize the light and heat resources, thereby establishing high-yield systems in the NCP (Meng et al., 2018b; Ren et al., 2021). Hence this practice was conducive to dietary energy production and economic-value creation. Most surveyed farmers in Quzhou used modern maize cultivars, but still adopted conventional planting rates (6–7 plants/m²) fitting outdated cultivars which were prone to lodging in a dense population. This knowledge gap needs to be filled to facilitate the farmers' adoption of this productive practice.

As for nutrient management, the lower N input for wheat (204 kg N/ha) and proper manure use (partial substitution of chemical fertilizer at the optimal N rate) were observed within exemplary systems (Table S10). Farmers in Quzhou tend to overuse N fertilizer for wheat (286 N/ha/year on average) as insurance against nutrient deficiency and yield loss. The wheat N rate of exemplary systems was closer to grain N removal of the high-yield wheat system (165 kg N/ha; Chuan et al., 2013), thereby avoiding massive N loss and GHG emission without compromising crop yield. Of fourteen wheat-maize cases with manure amendment, most non-exemplary systems overused inorganic N fertilizer (250–660 kg N/ha, Table S10) thus offsetting the beneficial effect of manure input (e.g., GHG emission mitigation). Manure should be used without exceeding the agronomically optimal N rate to provide the desired environmental benefits (Zhang et al., 2020), as the only one

exemplary system applying manure did. Otherwise, manure input would lead to extra N loss and economic and labor costs.

Besides, exemplary systems had a smaller number of irrigation events (3 and 1 for wheat and maize respectively) by canceling irrigation events at some non-water-critical growth stages (e.g., the regreening stage of wheat and some precipitation-dense periods during the maize season). This practice could not only mitigate groundwater depletion, but also decrease labor use and electricity-incurred economic costs, and GHG emission, without taking a toll on agronomic output. Previous studies showed that rationally reduced irrigation stimulated root development to enhance water uptake, and extended crop grain-filling duration, thus maintained or increased grain yield in wheat-maize systems, which supported our findings (Xu et al., 2016; Wang et al., 2018).

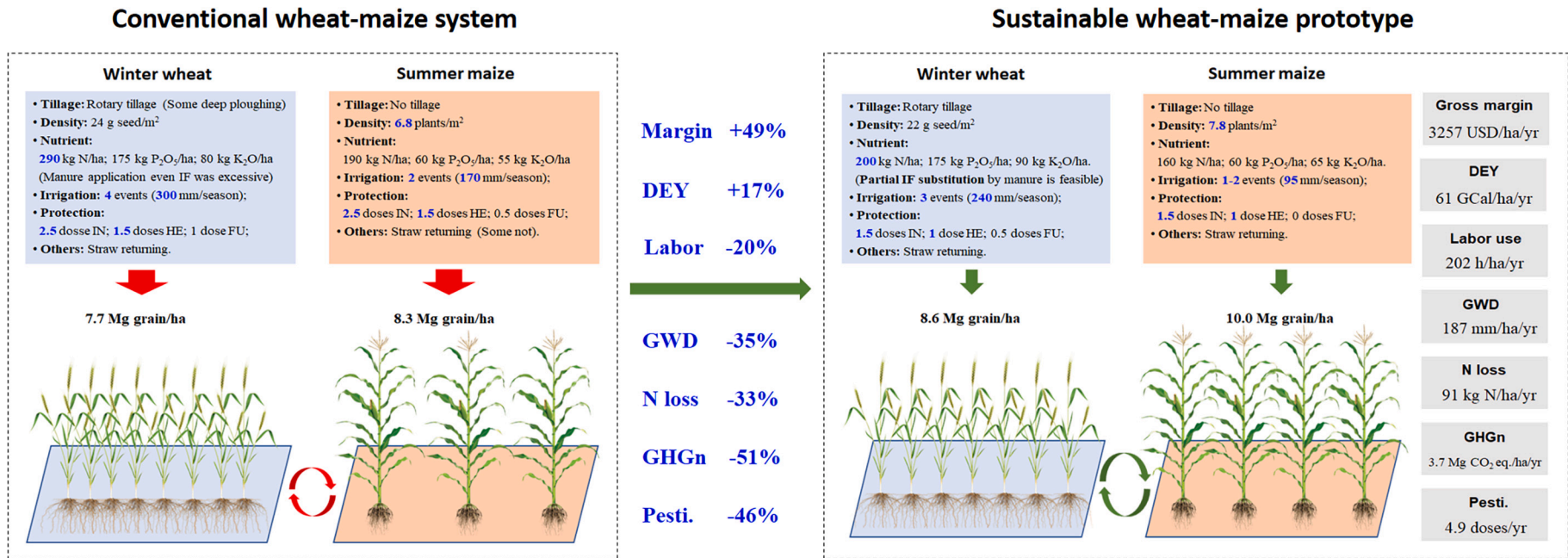
Exemplary systems also had fewer insecticide and herbicide applications than other systems, which might be related to a selection of better targeted pesticide types, appropriate application timings, and moderate nutrient and water input. Misuse of chemical pesticides was widespread in crop production in Quzhou. For example, repeated use of the same insecticide products, spraying herbicides later than best timings (like after weed outbreak), and combating target pests or weeds with wrong pesticide products, have resulted in overuse and sometimes crop damage (Li et al., 2014; Zhang et al., 2015). Besides, optimal nutrient and water input could promote plant quality and growth, thus increasing the resistance to pests (Médiène et al., 2011) and competitiveness to weeds (Kaur et al., 2018), which could reduce the reliance on external pesticides. Insecticide and herbicide use in exemplary systems needs to be further investigated to assess the scope for more efficient crop protection in Quzhou.

There were no statistical differences in biophysical and socioeconomic context between exemplary and other systems (Table 2). The soil dataset showed that the range of soil properties across Quzhou was narrow (Li et al., 2022). For instance, the range of soil organic matter content of wheat-maize fields was 1–2%, which would not be enough to affect crop growth under the high-level nutrient and water inputs (Oelofse et al., 2015). The variability of household backgrounds of exemplary systems was large, indicating those exemplary systems were developed by farmers in contrasting socioeconomic farm situations. This outcome justified the assumption that the identified exemplary systems would be transferrable and viable for wider adoption in Quzhou.

4.3. Development and future implementation of the sustainable prototype

In this study, a sustainable prototype for wheat-maize production was developed to facilitate the promotion of identified exemplary systems (Fig. 6). This prototype represented the mean values of nine exemplary cases since the variation among these cases was small, given the wide range of performance across the entire set of wheat-maize cases. Manure substitution, characteristic for one of the exemplary cases, was also captured as a flexible option in the prototype. After that, we discussed our prototype with several local agronomists and they ascertained its viability based on their knowledge and experience. If, in contrast to our findings, the variation within the selected cluster would be so large that cases in the cluster could not be aggregated into a certain type of cropping system, researchers could propose multiple sustainable prototypes that correspond to individual exemplary cases.

In further work, it is needed to work with a population of farmers with sub-optimal wheat-maize systems to adopt the practices in the sustainable prototype and to verify that their performance is improved as expected. We could provide technology training on this prototype to participant farmers and organize these farmers to visit households with exemplary systems. The implementation of this prototype would be easily accepted by farmers because it was locally proven to help gain more farming incomes. The local government could also offer subsidies to participants in order to further incentivize wider promotion as this prototype could greatly lower environmental costs, thereby improving the public benefits. Moreover, if some sub-optimal systems improve



their performance by using identified sustainable practices, there would be new Pareto-optimal and exemplary wheat-maize cases. In this situation, repeated application of the positive deviance approach for cropping system redesign might help the local whole set of cropping systems to move in the desired direction in an evolutionary manner.

4.4. Limitation and outlook

Uncertainty exists in the presented multi-criteria assessment because our indicators were empirically quantified using many parameters, especially in the environmental domain. To minimize the effects of result uncertainty, we used well-accepted methods to calculate the indicators (groundwater depletion: Yang et al., 2021; N loss: Cui et al., 2018; net GHG emission: Xu and Lan, 2017; pesticide use: Lechenet et al., 2017), and then extracted parameters calibrated to local conditions from a range of publications, statistics, and expert knowledge (see Table S2-S6). We also compared our empirical results with those measured in local experiments. For example, the estimated amount of soil C stock change (soil C sequestration minus degradation) was close to previous in situ measurement results (Cui et al., 2022). The average groundwater depletion and N loss in this study were also comparable to the results from field trials in the NCP (Yang et al., 2015; Song et al., 2018; Sha et al., 2020; Zheng et al., 2020). This indicates that our multi-criteria assessment is reasonable and reliable, although not as accurate as on-site measurement data. Measurement data could be included in future analysis to improve result certainty if available.

We considered as explanatory variables the most important management variables related to wheat-maize performance. However, some management practices were still not captured due to data accessibility, e.g., farmers did not remember the name of cultivars they used, and neither could they give us exact sowing and harvest dates. During our survey work, we found that the variety selection (mostly Jimai 22 and Luyuan 502 for wheat; Denghai 605 for maize) and cropping calendar (Oct.5-Oct.10 and Jun.1–10 for wheat and maize sowing respectively) were quite homogenous among wheat-maize observations. Therefore, variation in varieties or sowing dates might not markedly affect cropping system performance. In future research, a tracking survey that monitors all cropping activities implemented during the whole cropping system duration could be an effective way to avoid missing information that farmers cannot precisely report in recall (Fumagalli et al., 2011).

Similarly, some contextual variables that were not recorded in this study might have affected the sustainability performance of the wheat-maize systems. For instance, access to technology training and to sources of animal manure could play a role. These unobserved farm characteristics would affect farmers' decisions on cultivation practices that are highly associated with cropping system performance, e.g., manure amendment. A broader range of variables regarding farming contexts could be considered in future study.

Exemplary systems could be further elaborated by conducting a return survey of farmers who practiced exemplary systems. It can be expected that the follow-up revisiting could help us gain more insights into their success, e.g., detailed information on crop management and farming context (Steinke et al., 2019), which could make the sustainable prototype more practical.

In the future, this nested positive deviance approach could also be applied in other major cropping systems elsewhere in which multiple sustainability challenges exist. Sustainability indicators and exemplary positive deviants can be selected in terms of the local reality and stakeholder preference. With a broader application of the positive deviance approach, global cropping systems would have the great potential to improve their sustainability, contributing to achieving the Sustainable Development Goals advocated by the United Nations.

5. Conclusion

We proposed a nested positive deviance approach to find feasible

entry-points to close multi-dimensional sustainability gaps of cropping systems. Taking wheat-maize double cropping in the NCP as a case, we empirically evaluated economic, societal, and environmental indicators relevant to local sustainability challenges, and identified exemplary systems that reached the best compromise across all indicators from all positive deviants. Then multifaceted sustainability gaps were quantified, and key factors explaining those gaps were determined. We found that the sustainability performance of wheat-maize systems greatly varied across households. Most positive deviants represented strong trade-offs among indicators, while nine cases could serve as exemplary systems due to their acceptable compromise among indicators. Results also suggested that sustainability gaps of wheat-maize systems could be closed if other farmers implemented the key management practices of exemplary systems. Those sustainable practices included increased maize planting density, moderate N input, lower number of irrigation events as well as insecticide and herbicide use. Meanwhile, no systematic differences in farming context were observed between exemplary systems and other cases. This study showed that the presented positive deviance approach is a pragmatic method to identify locally viable solutions to improve sustainability of cropping systems by deciphering the diversity of farmers' practices.

CRediT authorship contribution statement

Zhengyuan Liang: Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **Wopke van der Werf:** Supervision, Writing – review & editing. **Zhan Xu:** Investigation, Data curation. **Jiali Cheng:** Investigation, Data curation. **Chong Wang:** Data curation, Resources. **Wen-Feng Cong:** Supervision, Writing – review & editing, Funding acquisition. **Chaochun Zhang:** Supervision, Resources, Funding acquisition. **Fusuo Zhang:** Supervision, Resources, Project administration. **Jeroen C.J. Groot:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

None.

Data availability

Data and R script used in the analyses are available from the corresponding authors on reasonable request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.agsy.2022.103471>.

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